

Haris Kamal Rana

Computer Science Undergraduate

Digital Twin Systems | Automotive Intelligence | Real-Time Backend Architecture

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EDUCATION

The University of Lahore

Lahore, Pakistan

Bachelor of Science in Computer Science (BSCS), CGPA: 3.66/4.00

Sep. 2023 – Expected Jun. 2027

- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Operating Systems, Object-Oriented Programming, Software Engineering, Computer Networks, Artificial Intelligence

ENGINEERING EXPERIENCE

DriveVitals – Founder & Lead Software Engineer | [GitHub](#)

Jul. 2026 – Present

Independent Engineering Project – Digital Twin & Fleet Intelligence Platform

Remote

- Architecting **DriveVitals**, a digital twin and fleet intelligence platform simulating commercial vehicle telemetry, driver behavior, and real-time fleet operations for predictive, decision-ready analytics.
- Designed a modular backend in **Python, FastAPI, WebSockets, and PostgreSQL**, delivering real-time telemetry streaming, fleet management, and analytics services consumed by a React-based monitoring dashboard.
- Built the core digital twin runtime – simulation orchestration, vehicle control, driver decision-making, physics-based vehicle behavior, and telemetry generation – as domain-driven backend services.
- Applied **SOLID principles**, dependency injection, and modular component design to keep the architecture extensible for future OBD-II telemetry integration and machine learning-based predictive maintenance.
- Lead a cross-functional team of three developers: defined project milestones, enforced a feature-branch Git workflow, and reviewed pull requests for controlled integration into the development branch.
- Designed the relational data model for fleet operations, historical telemetry, trip management, driver performance, and vehicle health analytics, engineered for extensibility toward predictive maintenance.

SELECTED ENGINEERING PROJECTS

AI Knowledge Platform | [GitHub](#) | Python, Streamlit, Groq API, RAG, Vector Search

2026

- Built a Retrieval-Augmented Generation (RAG) system that answers engineering questions from a personal Obsidian knowledge base, grounding responses in real project history rather than generic model priors.
- Developed an indexing pipeline with document chunking, embeddings, and vector search to provide context-aware responses through the Groq API.
- Implemented an automated GitHub commit watcher that converts development activity into structured engineering notes and synchronizes them with the Obsidian vault, keeping the knowledge base self-updating.
- Designed a modular architecture separating document ingestion, indexing, retrieval, and response generation for future multi-model extensibility.

Smart Door Security System | [GitHub](#) | Python, OpenCV, Raspberry Pi, Computer Vision

2025

- Built an embedded access-control system using Python, OpenCV, and Raspberry Pi for real-time facial authentication, optimized for CPU-only asynchronous inference with frame-skipping.
- Implemented a computer vision pipeline combining face detection, feature encoding, facial recognition, and access decision logic.
- Developed head pose-based liveness verification with randomized prompts to defeat spoofing attempts and improve authentication reliability.
- Integrated GPIO-based door control with PIN fallback authentication and event logging for secure physical access.

Vehicle Service Workshop Management Database | [GitHub](#) | SQL, MySQL, Database Design

2026

- Designed a normalized relational database for managing customers, vehicles, repair orders, mechanics, inventory, and payment records.
- Applied normalization (1NF–3NF), foreign-key constraints, and indexing strategies to ensure data integrity and optimize query performance.
- Developed SQL queries, views, and reports supporting service history, workshop operations, and business analytics.

TECHNICAL SKILLS

Core Engineering: Python, C++, Java, JavaScript, TypeScript, SQL, Data Structures & Algorithms, Object-Oriented Programming, System Design

Intelligent Systems: Retrieval-Augmented Generation (RAG), LLM Integration, Vector Search, Computer Vision, Digital Twins, Vehicle Telemetry, OpenCV, NumPy, Pandas

Infrastructure: PostgreSQL, MySQL, Database Design, Relational Modeling, Docker, Linux, Raspberry Pi

Software Engineering: FastAPI, React, Streamlit, SQLAlchemy, Pydantic, REST APIs, WebSockets, SOLID Principles, Design Patterns, Dependency Injection, Git, GitHub, Agile Development

CERTIFICATIONS

Elements of AI

University of Helsinki & MinnaLearn

Jul. 2025

Credential